

Mathematics & Statistics Roadmap for AI/ML

1. Linear Algebra

- 1 Scalars, Vectors, Matrices, Tensors
- 2 Matrix operations (Addition, Multiplication, Transpose, Inverse)
- 3 Dot product & Cross product
- 4 Eigenvalues & Eigenvectors
- 5 Diagonalization
- 6 Singular Value Decomposition (SVD)
- 7 Rank of matrix
- 8 Linear independence

2. Calculus

- 1 Functions & Limits
- 2 Derivatives
- 3 Partial derivatives
- 4 Chain rule
- 5 Gradient & Gradient Descent
- 6 Multivariable calculus
- 7 Jacobian & Hessian matrix
- 8 Optimization techniques

3. Probability Theory

- 1 Random variables
- 2 Probability rules
- 3 Conditional probability
- 4 Bayes Theorem
- 5 Probability distributions (Normal, Bernoulli, Binomial)
- 6 Expectation & Variance

4. Statistics

- 1 Mean, Median, Mode
- 2 Variance & Standard Deviation
- 3 Skewness & Kurtosis
- 4 Hypothesis testing
- 5 Confidence intervals
- 6 p-value

5. Optimization Techniques

- 1 Gradient Descent (Batch, Stochastic, Mini-batch)
- 2 Adam optimizer
- 3 RMSProp

- 4 Momentum

6. Discrete Mathematics

- 1 Logic & Boolean algebra
- 2 Graph theory basics
- 3 Combinatorics

7. Information Theory

- 1 Entropy
- 2 Cross-Entropy
- 3 KL Divergence

8. Numerical Methods

- 1 Numerical optimization
- 2 Floating point precision
- 3 Iterative methods

9. Graph Theory

- 1 Nodes & edges
- 2 Graph traversal
- 3 Shortest path algorithms